

**“From policy to practice:
How the British railway industry uses the PACE
framework for requirements management”**

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About me

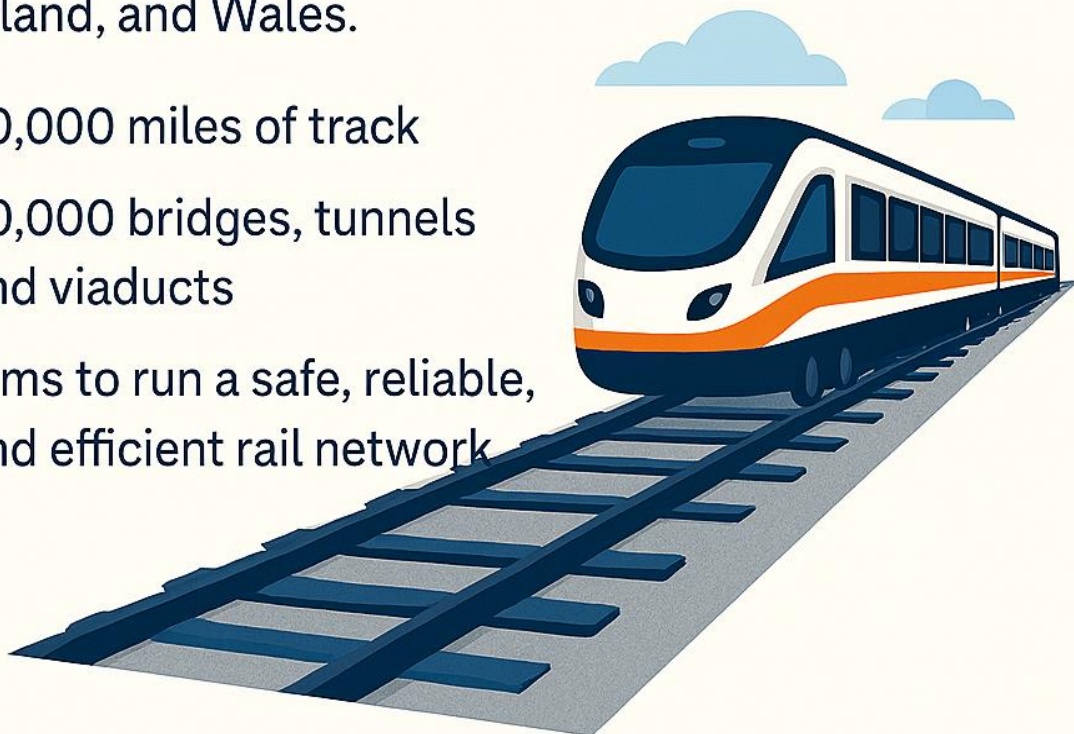
- Bus Driver
- Engineering Manager (Maintenance) for National Express Bus
- Operations Manager for the Police Service (Olympics 2012 and NATO 2014)
- Senior Requirements Engineer for Network Rail NW&C Region
- Systems Engineer for California High Speed Rail (Network Rail Consulting)
- Senior Project Engineer for Northern Powerhouse Rail
- IET Volunteer for the Railway Technical Network
- Volunteer for the Railway Industry Neuroinclusive Committee (RINC)
- University of Birmingham UK volunteer – guest lecturer and new students induction
- Technology University of Warszawa, Poland – guest lecturer

About Network Rail

Network Rail

Network Rail owns, operates, maintains, and develops railway infrastructure across England, Scotland, and Wales.

- 20,000 miles of track
- 30,000 bridges, tunnels and viaducts
- Aims to run a safe, reliable, and efficient rail network



It is a public sector, government-owned body with no shareholders; all income is reinvested into the railway.

Network Rail is the infrastructure manager. Train services are provided by Train Operating Companies (TOCs) and Freight Operating Companies (FOCs), who use the infrastructure it maintains.

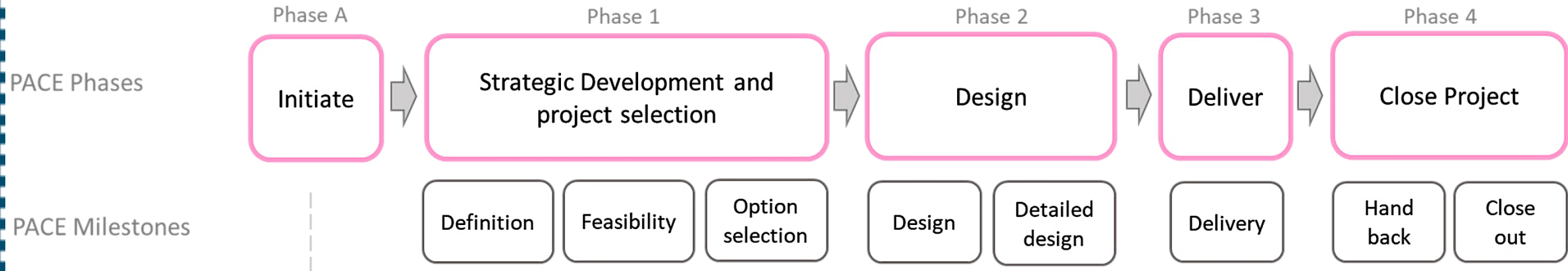
Network Rail operates in 5 geographical regions:

- Scotland's Railway
- North West & Central (NW&C)
 - Eastern
- Wales & Western
 - Southern

About PACE

Project Acceleration in a Controlled Environment

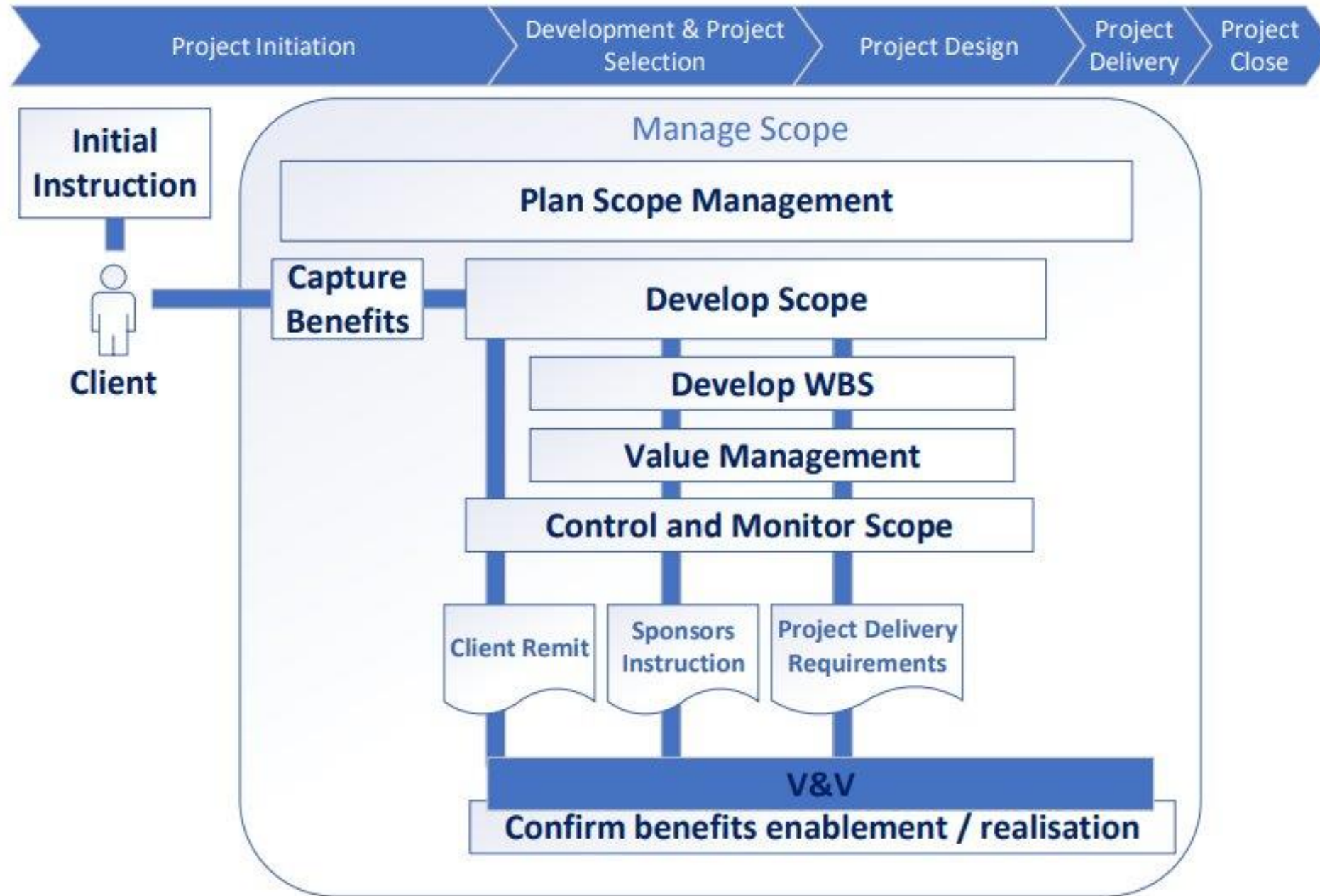
PACE is Network Rail's governance and project delivery framework for capital investment - covering projects, programmes, and portfolios. It provides a structured lifecycle, milestones, and assurance controls, and includes a comprehensive Product Index of required deliverables.



Replacing the older GRIP (Guidance for Railway Investment Projects) framework in 2021, PACE was introduced to enable faster decision-making, fewer sequential gates, and more flexibility in how teams deliver projects. Under GRIP's rigid eight-stage process, phase-by-phase sign-off often slowed delivery; by contrast, PACE allows phases to overlap, delegating decision authority to project teams and reducing delays.

Requirements Management under PACE





About this presentation

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**“An investigation into challenges with
application of Requirements
Management processes at Network Rail
(2023)”**

Research objectives

Define:

- Barriers in the current RM process
- Categorise issues with RM
- Analyse suitability of PACE (Manage Scope) framework
- Impact of change
- Risks
- Opportunities
- Recommendations

Investigate:

- Current RM practices
- Analysis of failures in RM
- Literature review
- Case studies
- Staff survey
- Quantitative and qualitative analysis of researched data

Research method

- Review of the state reports of the railway industry between 2011 and 2021,
- Analysis of the organisational structure and the effects of recent changes,
- Evaluation of the historic investigation into challenges with consistent RM adoption,
- Case study: costly and risky impact of scope changes with the effects of human perception of “it’s just a minor change”,
- Data research into inconsistencies with application of RM principles captured in the internal Library of Lessons Learnt,
- Further analysis of the lessons learnt findings:
 - Development of the Process Failure Mode Effects Analysis,
 - Comparison of the findings with the results of the CSH evaluation,
 - Study of the failure introduction and failure detection to determine highest risks in efficient project delivery,
 - Further analysis of data using IDEF0 modelling technique for assessing continuous improvement in the RM process,
- Completion of the Network Rail survey into user views on the RM process, its functionality, useability and applicability, and the qualitative analysis of the survey findings,
- Review of the cultural survey results into engineering competencies and learning appetite,
- Analysis of the reviews submitted by the participants of a local upskilling initiative: “Requirements Management Masterclass”,
- Review of the examples of catastrophic results of RM problems.

Literature review

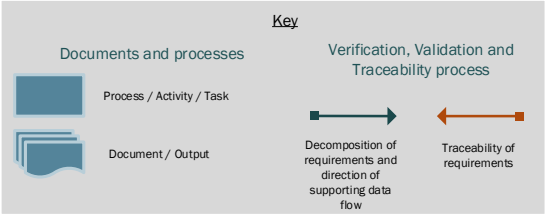
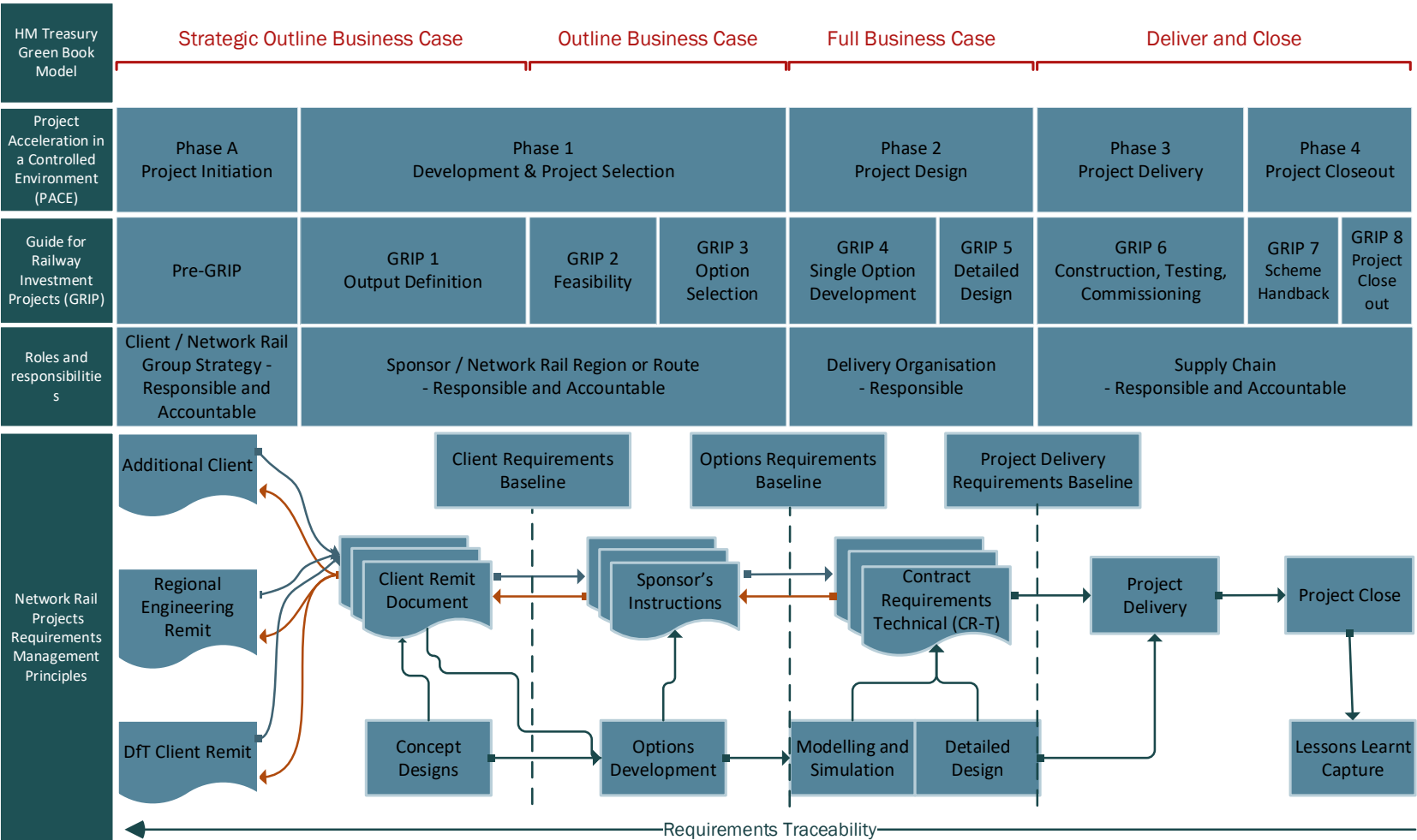
- Focus on high level factors influencing the railway industry - funding, project delivery, management and engineering principles for efficient and effective project delivery.
- Benefits of adoption of Systems Engineering and Requirements Management techniques.
- Challenges with Requirements Management including examples of catastrophic consequences of unmanaged RM issues.
- The need to distinguish process and human input.

Requirements Management in theory

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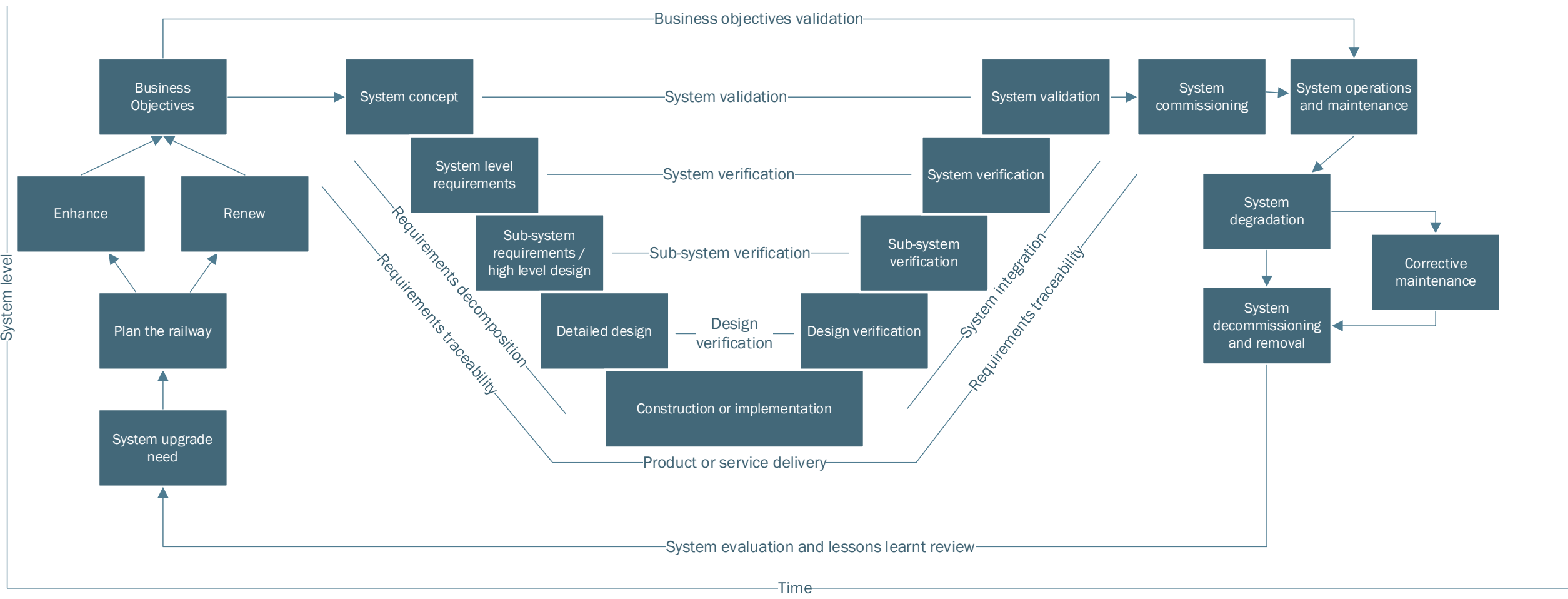


- Issues with requirements are common and expensive.
- Importance of alignment with SE principles.
- Network Rail managing requirements using PACE framework.
- PACE alignment with the HM Treasury Green Book to ensure legal, safety and management compliance.



My version of the railway specific V lifecycle model

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Soft Systems Methodology

To address the objectives of this project, it was important to distinguish (and then re-connect) process and people in a hypothetical observation:

- a) Process and methodology are control tools designed to provide structure and direction of a project, where its output is to be delivered by b) within set timescales and budget; and
- b) People, whilst their skills, knowledge and competencies are fundamental to deliver a), individually (and in groups) they present different values, views, and perceptions.

RM, as a working method can be challenging, costly, and commonly difficult if arising problems are unmanaged or unmaintained.

One would say that there's always room for improvement, but the word *improvement* in RM holds an unambiguous meaning. This is because improvements constitute of different matters based on personal perception and understanding of the problem, hence its close relation to- and dependency on the Soft Systems Methodology (SSM).

Preliminary investigations

- Results of a historic investigation into challenges with RM adoption showing recurring and unresolved issues.

Table 4 NR Internal investigation into gaps in requirements management process implementation
(Krusinowska, 2016)

Author and publish date	Source	Summary of findings
Network Rail, London North Western Route, October 2016	Internal investigation: "Gaps in Requirements Management process implementation – selected LNW Schemes" (Krusinowska, 2016)	▪ Ambiguous NR standards, lack of consultation with stakeholders to avoid misinterpretation. ▪ Route Asset Manager (RAM) not involved or consulted post GRIP4. ▪ Deliverer managing subcontractors – no visibility to RAM. ▪ Late project scope changes not consulted or communicated with adequate stakeholders. ▪ Gaps in communication between Deliverer and the Route/Client. ▪ DOORS system (RM recommended repository) not used - seen as too complicated to operate. Number of different document storage systems used across the business resulting in missing records. ▪ Not all projects are aware of National Records Group (NRG) function. ▪ Records not returned to NRG by project team/deliverer/contractor. ▪ Stakeholders not identified in early project stages. ▪ Deliverer not involved in early project stages. ▪ Deliverer requires more awareness of RM process. ▪ RM not fully integrated with GRIP. ▪ Deliverer (contractor) failing to hand back projects. ▪ Value Engineering – due to cost limits requirements are prioritised or/and removed. ▪ NR commercial strategy inefficient.

Preliminary investigations

- Results of a historic investigation into challenges with RM adoption showing recurring and unresolved issues
- Effects of devolution and removal of centralised RM support function/department.

Until Q1 2022, the Rail Investment Centre of Excellence (RiCoE) department of NR hosted a small team of requirements managers dedicated to owning and managing NR requirements policy, as well as supporting programmes and projects across the UK with application of relevant standards, enhancing skills and providing centralised support platform, including internally managed web resource.

The requirements management team was disbanded due to organisational restructuring in Q1 2022 and due to devolution, which “pushed” responsibility to regions to own requirements management process.

The ownership of the PACE standard and provision of guidance and assurance of the capital portfolio remained within the RiCoE however, the importance of maintaining national focus on RM diminished, affecting NR and its supply chain and partners.

RISK = OPPORTUNITY

Preliminary investigations

- Effects of devolution and removal of centralised RM support function/department
- Results of a historic investigation into challenges with RM adoption showing recurring and unresolved issues
- Cultural survey into engineering competencies revealing gaps in project delivery knowledge and appetite for learning about SE, RM and integration.

The NW&C Enhancements and Renewals Engineering (ERE) Cultural Survey was remitted in 2021 to determine levels of confidence in managing engineering elements of infrastructure projects. Respondents were from a range of project engineering and construction management roles, 32% of whom were from the Construction Management discipline.

Results showed that 40% of respondents did not know what their Project Level of Control (LoC) was (now PACE Project Complexity Assessment or PCA).

Survey responses show a trend in a need for knowledge advancement in the areas of SE, contracts and procurement, and RM.

The survey identified significant interest in areas of railway interoperability and common safety method for risk evaluation and assessment (CSM-RA).

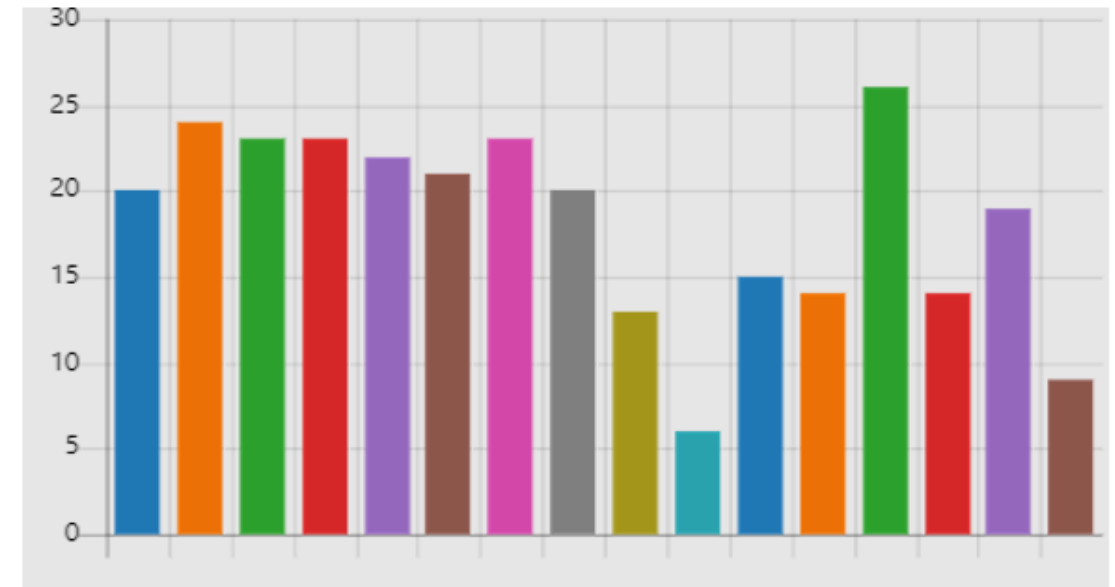
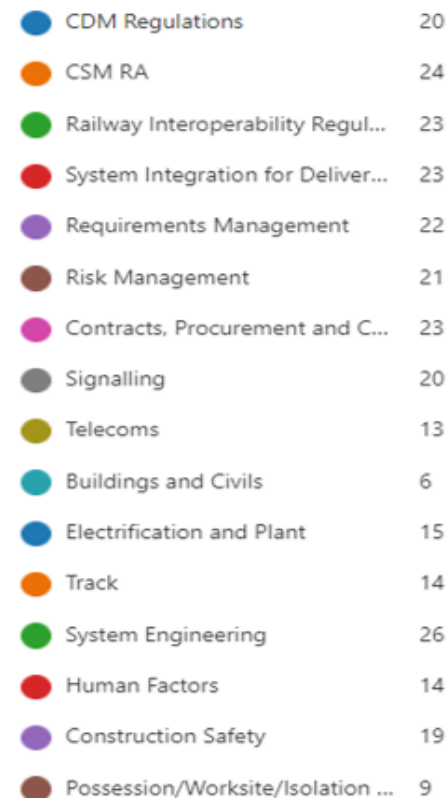


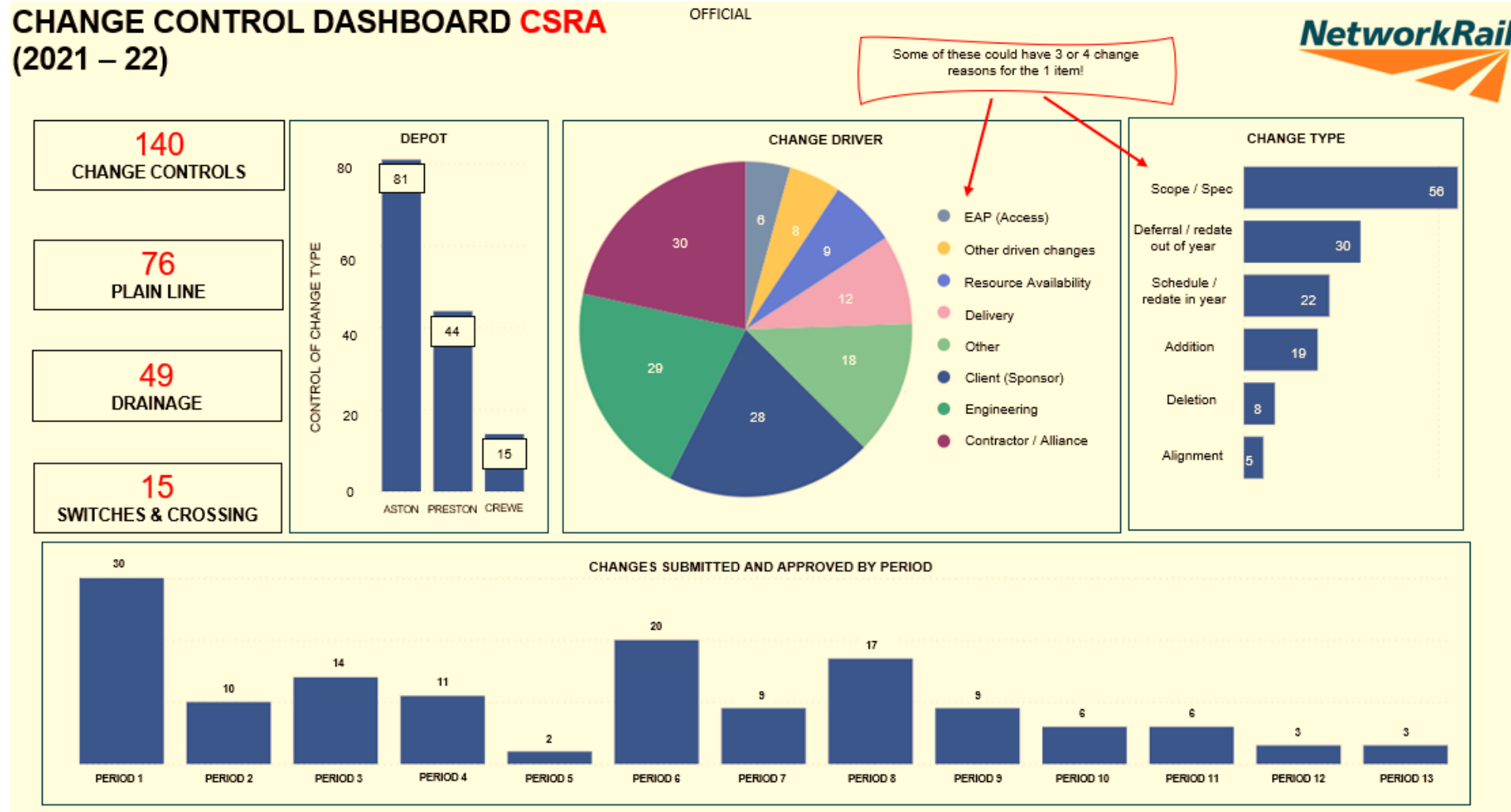
Figure 5 NR internal Cultural Survey 2021 results

Preliminary investigations

- Effects of devolution and removal of centralised RM support function/department
- Results of a historic investigation into challenges with RM adoption showing recurring and unresolved issues
- Cultural survey into engineering competencies revealing gaps in project delivery knowledge and appetite for learning about SE, RM and integration
- Costly and risky impact of change – study into common working practices affecting project including perception of “just a minor change”.

The impact of “simple” change in requirements on a railway project (add wet bed removal to the original scope) – NW&C WD Track Renewals team.

NW&C WDTR team reports receiving on average 23 change requests per period, with 56 changes associated with project scope in a year between April 2021 and March 2022.



Activity	Time required to complete the activity (in PD – person day)
New or amendment to existing survey, and validation of existing measurements, timescales, costs and running line possession requirements.	6 PDs
Re-design including delayed average financial costs, specification changes and approval processes.	3 PDs
Site walk out to ensure points, earthing points, cables and other elements are identified in the extended track section. Activity includes personnel from different asset engineering groups.	4 PDs
Revised database, dilapidation details and lifting plans.	1 PD
New wagon plan, ballast calculations, locomotive limits, train route, train weight, lifting plans and outputs.	3 PDs
Updates to material orders including on track machinery and train loads.	3 PDs
Revised cost stack in terms of commercial and financial changes.	2 PDs
Revisited project stage gates including reporting elements.	3 PDs
Raise of change control process including costing, approval, records, updates, and relevant system input.	2 PDs
Total personnel days required to prepare for new, changed activity:	27 PDs

27 person days x 56 project scope changes = 1512 person days

~£250 per person per day x 1512 person days = **£378,000**

My own bias

The need for a self-regulating view on the issues with RM at NR through establishing boundary judgements, and categorising and answering relevant “is” and “ought” questions based on Critical System Heuristics method by Ulrich (2000).

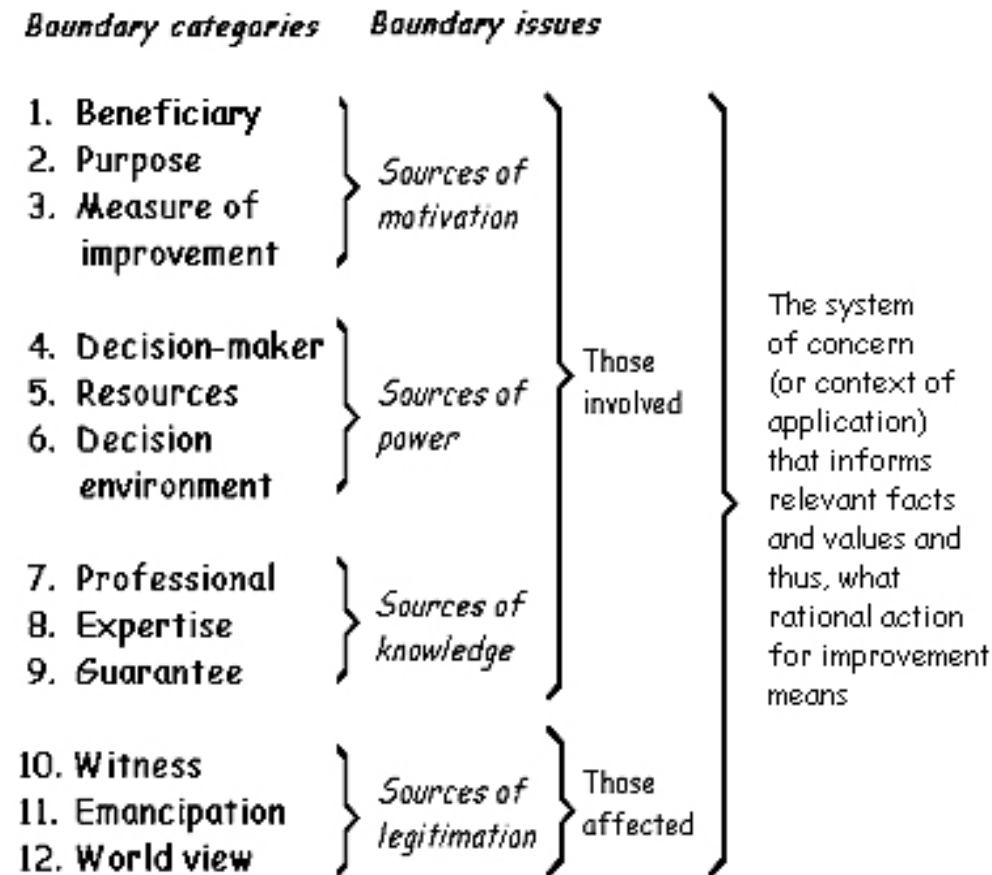


Table 8 Identified lessons learnt categories and indicative root causes of insufficient application of requirements management process

Case studies

Network Rail Lessons Learnt Library:

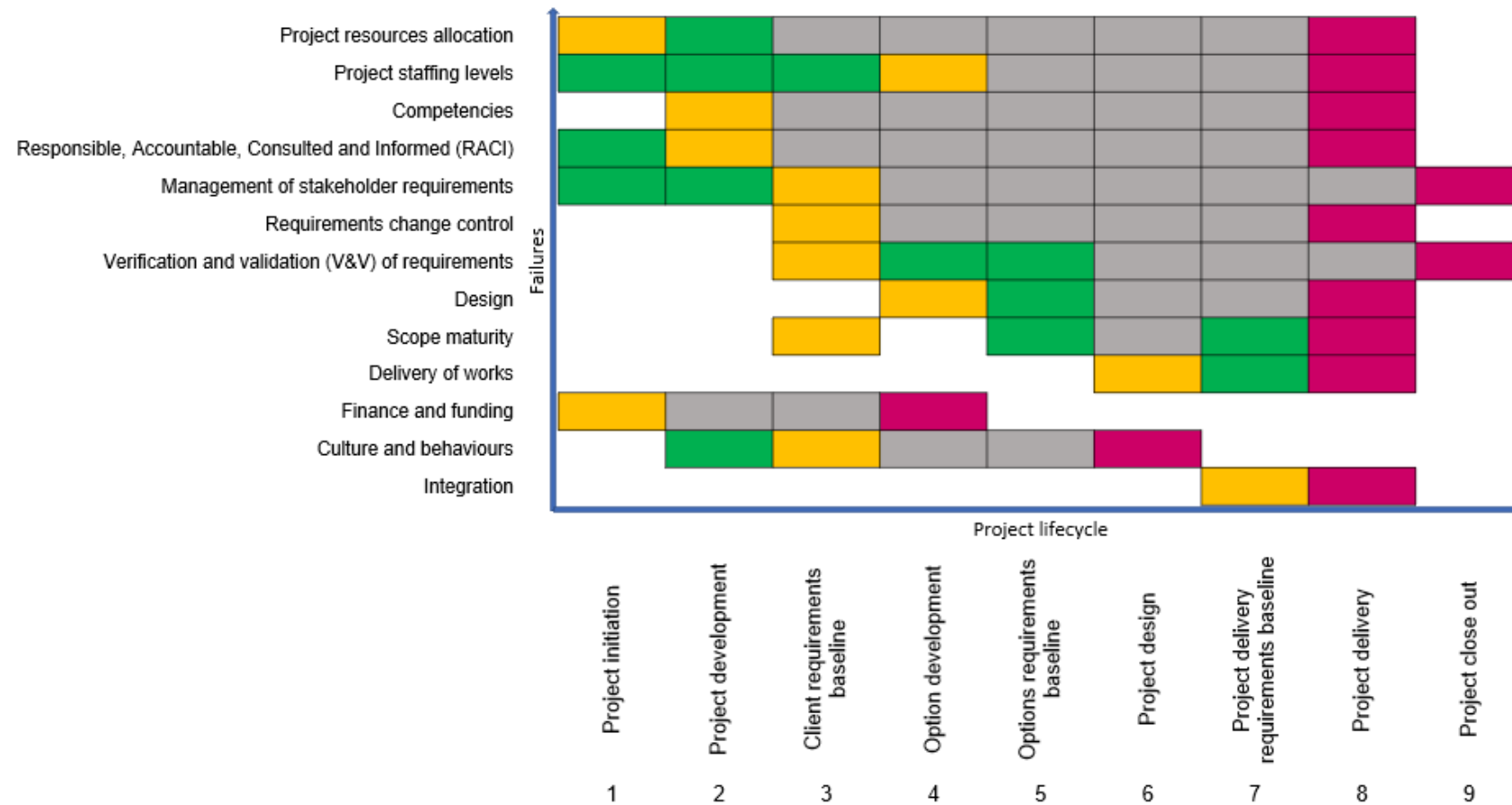
- Reliable source of project data.
- Thousands of entries related to requirements management.
- Findings were categorised to identify and evaluate problematic areas (top 8):

Potential root cause - category	Description
Roles and responsibilities	Insufficient resources, high staff turnover and misallocation of designated personnel.
Stakeholder engagement	Problems related to lack of, or weak level of consultation with relevant requirements providers, and absence of structured engagement in capturing requirements.
Design	Incomplete designs and lack of sufficient data before commencement of construction or building work, misinterpretation of NR standards and calculations by third party designers.
Scope maturity	Inadequate quality of scope, delays in reaching scope agreement, late changes, and unclear requirements. Evidence related to disagreements between critical stakeholders in building requirements set, lack of control of volume and quality of requirements.
Delivery of works	Insufficient construction methodologies, lack of assurance during construction works towards design errors, and unplanned and miscommunicated changes in materials used.
Finance	Increased costs due to (late) changes to the scope, misidentification of funding streams, lack of survey data set, and late redesign.
Culture	Ineffective leadership in challenging situations, mismanaged project risk indicators, miscommunication, absence of personal interest in project delivery, and disjointed approach.
Integration	Insufficient appreciation or understanding of other engineering disciplines, failures in identifying and integrating corresponding planned work and surrounding environment.

(Process) Failure Mode Effects Analysis:

- Pre-categorised functions examined in the context of failure modes, their effects, potential reasons for further establishment of severity, probability and detection factors.
- Risk Priority Numbers indicate four high-risk areas impacting directly on passengers and quality of railway customer experience:
 - Management of stakeholder requirements;
 - Scope maturity;
 - Project staffing levels;
 - Competencies.
- Findings consistent with CSH evaluation results.

Failure introduction and failure detection study reveals gaps where risks are most likely to be affecting projects:



Key

introduction

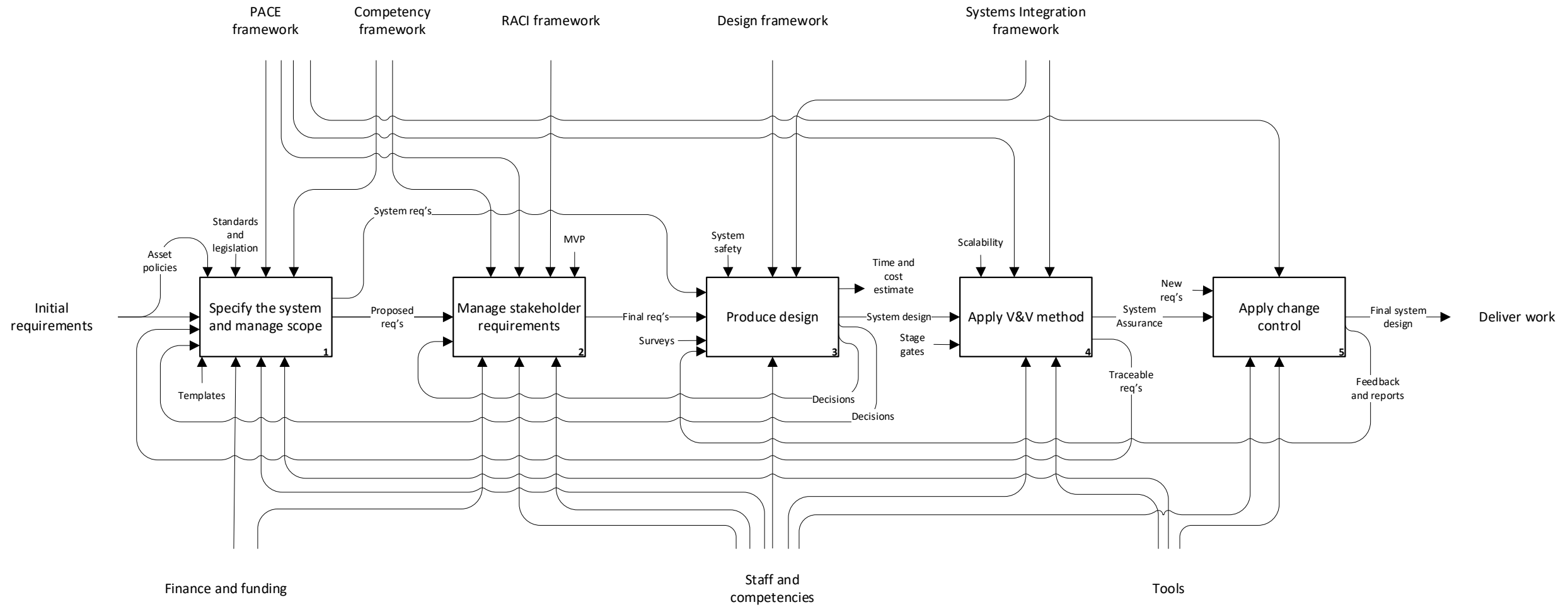
mitigation

introduction and mitigation

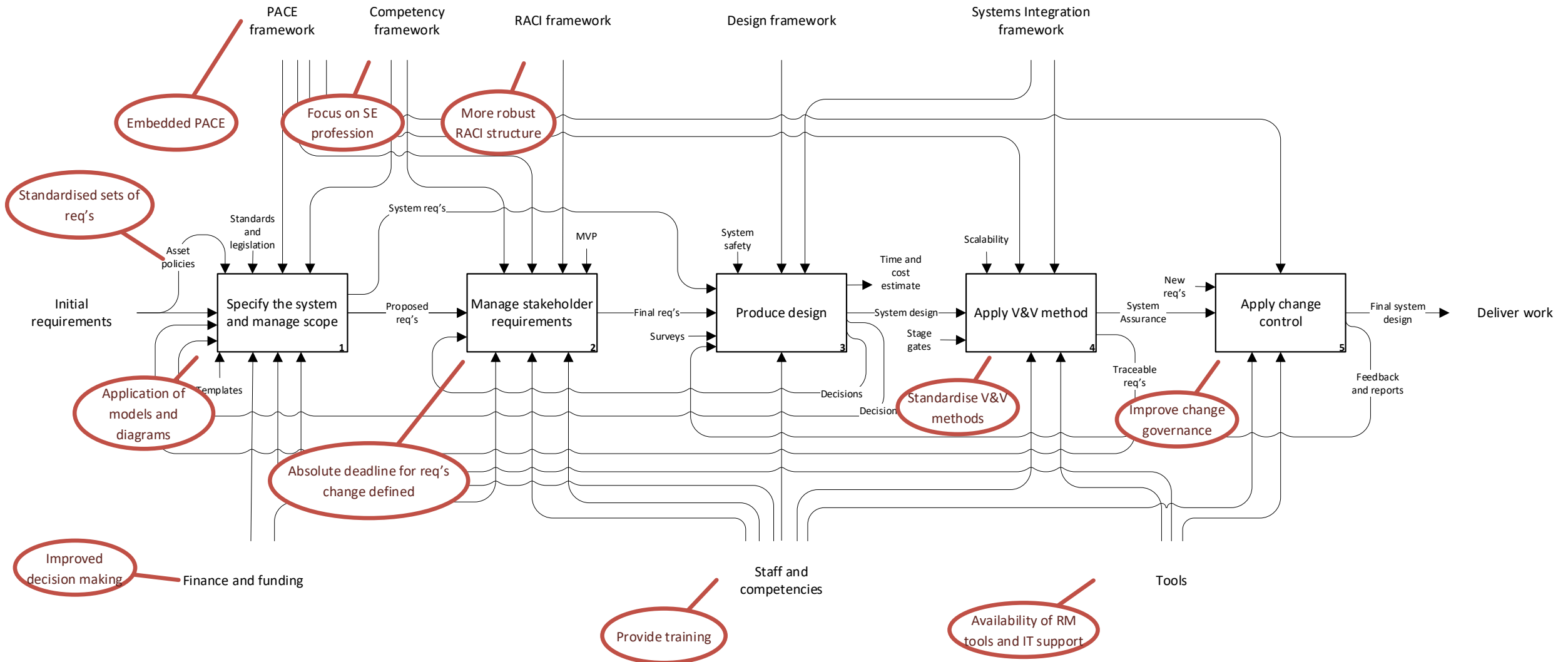
risk area

detection

IDEF0: modelling technique for assessing continuous improvement in the RM process:



IDEF0: modelling technique for assessing continuous improvement in the RM process **and for providing improvement recommendations.**



Network Rail Requirements Management survey

Table 11 Network Rail-wide survey into requirements management process - collective responses to questions 1, 2, 3, 4, 5, 6 and 8.

- Over 500 participants invited to take part, 62 responded.
- Qualitative data review to understand user views on RM process, its functionality, useability and applicability.
- Role-based responses helped capture concerns that are viewed as an opportunity to integrate RM activities but also to share with other groups who may not share the same approaches.

Q	Summary of answers: collectively
1 and 2	Participants included: 17 asset owners (majority dealing with requirements specification). 24 people representing project delivery (majority involved in remits, specifying requirements, and overall project lifecycle including assurance). 2 people were from sponsorship (involved in scheme delivery, overseeing the process and assurance). 12 were systems engineers and requirements managers (majority overseeing the whole RM process throughout project lifecycle). 7 people represented other, regional roles including functions, assurance, and strategy (overlooking all stages of project lifecycle).
3	Majority of responders gained their knowledge and skills in RM from self-study, hands-on experience and working with others. 20 out of 62 people have academic qualifications in SE or RM.
4	3 people don't have any confidence in RM skills. 4 people are experts. Others have various degrees of confidence.
5	32 people feel the RM process at NR is unclear. 26 people said the process is clear. 2 people didn't have enough knowledge about the process to comment. Majority said that the process is lacking scalability, consistency and understanding by others.
6	27 people felt confident in finding alternative tools, including the use of excel. 15 people didn't feel confident. 20 people said that the use of tools was not part of their role.
8	50 people can find information and help. 12 others unable to find up to date resources due to decentralised RM function.

Summary	
Asset owners	Asset owners' engagement adds most value at the early parts of project lifecycle where requirements specification is fundamental for further decomposition. Those professionals felt excluded from later stages of projects where iterations to requirements and late changes were introduced, but not consulted with them. They felt that communication from other parties is insufficient, which increases risks and non-compliance with standards, and reported that RM training and availability of information needs addressing.
Project deliverers	Project deliverers said that other working groups do not see value in managing requirements and are negligent or inconsistent towards the RM process stages, including late changes. Responders said that PACE is unclear and unscalable, interfaces are mismanaged, and project RACI isn't specified meaning that resources are inadequate to project scope.
Sponsors	Sponsors specified insufficient competencies and skillset as core problem, with added difficulties from late changes to scope, gold-plating requirements, and unspecified project RACI.
Systems Engineers and Requirements Managers	Systems engineers and requirements managers' opinions were based on processes and strongly indicated insufficiency in knowledge and application of systems engineering principles, including SI and RM practices across the organisation. Professionals felt that resources and tools are scarce, in-house training is not available, PACE is allowing too much flexibility where the process is not fully and consistently understood by others, and late changes are widely acceptable without restrictions.
Others	The group provided combined outlook on competencies, behaviours, and culture as major influencing factors in RM challenges that carry risks. They felt that PACE provides well-defined structure however, inconsistency and misunderstanding of it impacts on the quality of its application, including allowance for late change. The group noted training, tools and requirements database are required.

Summary of case studies findings

Case study	Findings
NR Lessons Learnt Library	1081 records were found to be associated with issues related to project delivery in the context of RM adoption. Reports were categorised and suggest the following areas of concern: roles and responsibilities, stakeholder engagement, design, scope maturity, delivery of works, finance, culture, and integration.
(P)FMEA	The study determined four flawed activities within RM: management of stakeholder requirements, scope maturity, project staffing levels, and competencies. Finance and funding were lowest scoring failures in the analysis, which is inconsistent with preliminary investigations, where impact of change resulted in excessive project overspend.
	The analysis of failure introduction and detection of failures identified in (P)FMEA table shows gaps in risk controls, where opportunities to mitigate project risks may be overlooked.
IDEFO modelling	IDEFO was used to capture five main functions in RM process where the flow of information and feedback loops show dependencies, constraints and outcomes used to suggest continuous improvement. The model allowed for identification of potential advancements in complex areas where SE principles could support the effectiveness of the process while reducing risks.
RM survey	62 participants responded to 12 questions exploring their knowledge, views and understanding of the RM process adopted by NR.
	Vast majority of survey participants were critical and challenging in their perceptions on the suitability and consistency in applying PACE framework, applicability of SE and RM principles, and behaviours of other project players.
	Staff reported concerns with project players working in isolation, NR not providing formal training in RM (including competency profiling), unavailability of NR-approved RM tools and absence of centralised support, including repository for RM-related information and guidance.

The summary of findings

- The British Government expresses the need for increased focus on efficiency and effectiveness in railway infrastructure projects delivery due to changing economy and passenger demand, while encouraging wider adaptation of the systems thinking by the industry.
- SE practices and systems thinking provide structure, governance and focus on risk reduction that supports RM process.
- Issues with RM are costly, common, and potentially catastrophic.
- SSM provides strong view on interdependence between process and human input based on behaviours, perceptions, and competencies.
- Devolution has unfavourable effects on RM adoption.
- It is recognised that upskilling on SE practices helps develop systems thinking approach in projects.
- “Simple” project alterations and scope changes result in significant cost increases.
- Identification of process failures and development of improvement ideas require prior identification and elimination of bias with the use of evaluation method, for example CSH.

Network Rail approach to RM - summary

- Following development of industry-wide SPEED principles, PACE framework is to *minimise bureaucracy and maximise value* however, knowledge and behaviours are insufficient to support this.
- Each Region is expected to manage requirements internally however, only NW&C and W&W are actively involved (at the time of publishing).
- Up-to-date training and upskilling on RM is not available nationally.
- Centralised RM support and up-to-date resources are not available.
- Overall high-level accountabilities and responsibilities are assigned to client, sponsor, delivery organisation and supply chain, but their confidence and technical knowledge in RM varies (mainly sponsors).
- Framework allows flexibility however, it is inconsistently understood or applied by all project players.
- Cooperation and integration are key to successful delivery of requirements, but silo working is too common.
- Personal perceptions of project players (SSM) are detrimental to the process components.
- Requirements Advisory Group (RAG) is not utilised to its potential.

Recommendations

- Project delivery framework PACE to include systems engineering principles adequate to the scale and type of projects.
- Increased national focus on RM with dedicated resource to drive initiatives nationally and to support Regions.
- Central repository for all requirements information.
- RM tools directory for projects to choose from (including demos, costs, useability, user experience, etc.).
- Increased application of system modelling techniques (+ education?) in RM activities.
- Development of Regional Requirements Management Strategy or Policy to declare ownership of the process for consistent, localised and tailored approach. To include dedicated representatives.
- Increase sponsors' confidence in RM.
- Increased collaboration with external partners to remove barriers in RM and improve decision-making.
- Increased national and local focus on and support to the Requirements Advisory Group.

Recommendations – further details

- National and/or Regional initiatives to increase knowledge and awareness of RM, including:
 - Provision of an up-to-date information platform including guidance and troubleshooting.
 - (Re)Development of RM training material.
 - Library of widely applicable, standardised requirements, including practical examples of RM scalability with templates and best practice.
 - Standardised RACI for variety of projects .
 - Further embedment of PACE and its modules in forms of briefings, workshops and Q&A.
 - Support in developing specific RM framework tailored for a region/route/portfolio or programme.
 - Potential reporting activities to monitor and act on requirements management gaps, issues, opportunities.

NW&C Requirements Management Masterclass as an example of good practice

- 20 masterclasses delivered with 247 attendees (including external colleagues) as of Feb 2024.
- 96% of attendees think the masterclass helps increase the confidence in writing/managing requirements.
- 97% of attendees would recommend the masterclass as part of professional development.

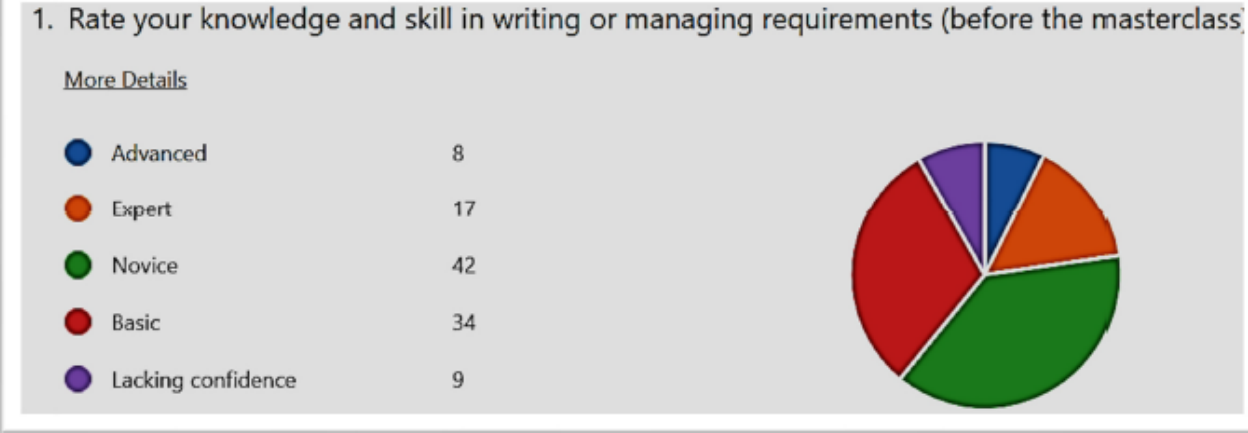


Figure 12 Requirements Management Masterclass - feedback indicating knowledge level before the session (Network Rail, 2022)

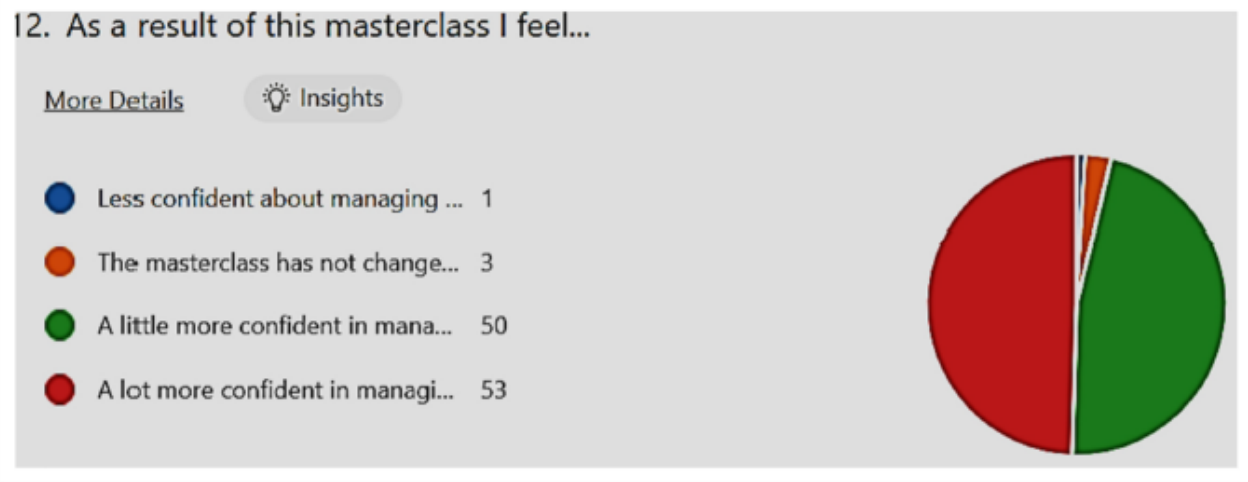


Figure 13 Requirements Management Masterclass - feedback indicating knowledge level after the session (Network Rail, 2022)

Locally developed and managed live library of assets and functions requirements (which are to be treated as a set of guide rails during project development), compatible with DOORS input and with potential to improve alignment with CR-Ts and support transversal requirements.

Further research suggestions

- Relationship between NR and its suppliers in terms of RM and investigation into suppliers' feedback of working with NR PACE framework.
- Effective communication and cooperation between external parties in managing requirements.
- The impact of innovation on RM in third party funded schemes.
- Funding and management of renewals work-bank in difficult financial periods driven by performance related targets and changing customer demand.
- RM tools and their suitability and useability in achieving project delivery efficiencies.
- Impact of AI on SE and RM.
- The future of SE and RM in light of Great British Rail (GBR).

Thank you

Questions?

